

“FINAL REPORT”

Energy Expenditure Estimation Machine Learning Model

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Abstract:

Energy expenditure estimation is one of the most essential tasks in health tracking, fitness tracking, and, in general, activity recognition systems in which activity awareness is involved. This study analyzes different machine learning algorithms and different variants of deep learning algorithms for measuring energy expenditure levels with smartphone-based accelerometers and gyro sensors in the UCI HAR dataset. As energy expenditure has a tight relationship with activity, activity recognition is considered as a surrogate in order to work with different intensity levels. Conventional machine learning algorithms, a fully connected network, as well as check RNN, LSTM, and GRU algorithms, are considered in this study. The experimental analysis demonstrates that all kinds of recurrent algorithms, especially GRU, outperform other algorithms, which asserts that time is a fundamental factor in expenditure patterns involved in motions.

1. Introduction/Problem Statement:

Energy Expenditure Estimation: It involves determining the energy used by a person in performing physical activities. It is important to estimate with accuracy in areas such as healthcare, obesity research, sports analysis, rehabilitation, and wearable computing.

Smartphones with installed accelerometers and gyroscopes are a low-cost platform for estimating energy expenditure through motion analysis. Activities such as walking, sitting, and standing etc. have varying energy expenditure; hence, activity-based estimation is appropriate.

1.1. Objective:

The goal is to infer the levels of energy expenditure by learning patterns of motion from smartphone sensor data based on the following:

- Feature-based machine learning
- Fully connected neural networks
- Recurrent neural networks for temporal modeling

2. Related Work:

From previous research, it has been shown that energy spent can be estimated using wearable sensors by means of activity classification and modeling of intensity. Traditional techniques rely on statistical feature engineering, while deep learning techniques, and more specifically recurrent architectures, were far more successful, exploiting the time-dependent characteristics of motion patterns.

3. Methodology:

3.1. Feature Engineering & Traditional Machine Learning:

Statistical, temporal, frequency domain-related features can now be extracted from the signal to measure the intensity of movement related to expenditure in energy.

Features extracted:

- **Statistics:**

mean, median, standard deviation, variance, min, max, skewness, kurtosis

- **Temporal:**

zero-crossing rate, signal energy, peak to peak amplitude, signal magnitude area

- **Frequency:**

Frequency-spectral centroid, spectral entropy

Models Used:

- Logistic Regression
- Support Vector Machine (RBF)
- Random Forest
- XGBoost

3.2. Fully Connected Neural Network

A fully connected neural network is then trained on these features to model non-linear patterns from motion features and different levels of energy expenditure. The network has various fully connected layers with ReLU activation functions, which is then optimized by Adam with cross-entropy loss.

3.3. Recurrent neural networks

To analyze the motion dynamics based on energy consumption, the Recurrent Neural Networks were directly trained on multivariable time-series data.

Models Evaluated:

These are the models we evaluated.

- RNN
- LSTM
- GRU

These models learn patterns in the movements of the data over time, and these learned patterns are significantly informative for the energy expenditure estimation.

4. Experiments and Results

4.1. Description of Dataset

UCI Human Activity Recognition HAR Dataset-Acceleration and gyroscopes data obtained by smartphones.

- Sensors:
Accelerometer and Gyroscope
- Activities :
Walking, Walking Upstairs, Walking Downstairs, Sitting, Standing, Laying
- Training samples: 7352
- Test samples: 2,947
- The output will be characterized by the following parameters:
- Sampling rate: 50Hz
- Sequence length: 128 timesteps

4.2. Data Settings

Train/Validation split: 80% / 20%

Preprocessing:

- time indexing
- Handling missing value
- Removing duplicates
- Feature scaling (for ML and FCNN)

Evaluation Metrics:

- Accuracy
- weighted F1-score

4.3. Results

Comparison of All Approaches

The following models have been implemented following the literature. The following table shows the results:

Model	Accuracy	F1-Score
Logistic Regression	61.59%	0.606
SVM (RBF)	60.84%	0.597
Random Forest	63.90%	0.636
XGBoost	64.85%	0.646
Fully Connected Neural Network	68.40%	0.673
RNN	36.85%	0.244
LSTM	21.75%	0.099
GRU	82.94%	0.827

4.4. Discussion

- Feature-based ML models are general in energy-related trends but are agnostic to time.
- FCNN enhances performance by capturing non-linear interactions of features.
- Instability of the gradient and Over fitting is observed in Vanilla RNN and LSTM.
- GRU is able to reach the highest accuracy, accurately abstracting the idea of temporal intensity of motion.
- Confusion between low-energy activities - Sitting vs Standing occurs due to patterns created from sensors being similar to each other.
- These observations validate the significance of temporal abstraction to achieve an accurate result to generate the estimated expenditure of energy.

5. Conclusion and Future Work

Conclusion

This work proves that the use of deep learning methods greatly assists in the calculation of energy expenditure from smartphone sensor data. Fair results are achieved by normal machine learning and FCNN algorithms. However, the most effective results are achieved by RNN algorithms, especially GRU algorithms because of their ability to model the motion dynamics.

Best Performing Model:

Recurrent Neural Network based on GRU. Its predicted accuracy is 82.94%.

Future Work

- Direct regression-based energy expenditure estimation (calories/METs)
- CNN–GRU hybrid architectures
- Deployment on real-time mobile systems